

Education

Institute of High Energy Physics, Beijing, China

Doctor of Philosophy, advisor: Qiang Zhao

March 2022 - June 2025

Theoretical Physics

Institute of High Energy Physics, Beijing, China

Master of Science

Sep 2019 - March 2022

Theoretical Physics

University of Chinese Academy of Sciences, Beijing, China

Bachelor of Science

Sep 2015 - June 2019

Department of Physics

Skills and Expertise

Language Chinese (native), English (proficient)

Programming Julia, Mathematica, Python, C++ , LaTeX etc

Interests and Skills Particle Phenomenology, Hadron spectroscopy, Effective field theories, S-matrix theory especially dispersion relations etc

Research Experience

Department of Physics, Old Dominion University

Post-Doctoral Research Associate

Virginia, US

Oct 2025 - present

- Theoretical study on hadronic/nuclear physics; S-matrix techniques on the study of hadron interactions; AI/ML applications in hadronic reactions.

Institute of High Energy Physics, Chinese Academy of Sciences

Ph.D student - Theoretical Physics

Beijing, China

Sep 2019 - June 2025

- Effective field theories, Scattering amplitudes and Dispersion relations
- Hadron spectroscopy, especially the exotics beyond quark model: hadronic molecules, hybrids, glueballs etc
- Final state interactions, especially kinematic singularities and three-body unitarity framework

Department of Physics, University of Chinese Academy of Sciences

Student Internship: Experiments and Data Analysis in High Energy Physics

Beijing, China

July 2018 - Jan 2019

- Ingredients of Particle physics, Collider Physics, Detector Physics etc
- Data analysis: Event selection, Multi-variable analysis, Boosted decision tree, RooFit etc

Talks And Teaching

Academic Talks

- The International Workshop on Partial Wave Analyses and Advanced Tools for Hadron Spectroscopy (PWA13/ATHOS8)
- The 23rd International Conference on Few-Body Problems in Physics (FB23)
- The 7th Symposium on “Symmetries and the emergence of Structure in QCD”
- 8 ~ 10 other domestic seminars or workshops

Teaching Assistant, University of Chinese Academy of Sciences

Feb 2024 – May 2024

- “An introduction to hadron physics” by Bingsong Zou, Qiang Zhao, Fengkun Guo
Quark model, hadron structure and form factors, QCD symmetries and EFTs (CHPTs and HQEFTs), S-matrix theory, dispersion relation, kinematics, hadron(-nucleon) interactions etc

Publication and Preprint

- **L.Qiu** and Q.Zhao, Towards the establishment of the light $J^{P(C)} = 1^{-(+)}$ hybrid nonet, CPC 46(2022)5,051001
- **L.Qiu**, C.Gong and Q.Zhao, Coupled-channel description of charmed heavy hadronic molecules within the meson-exchange model and its implication, PRD109(2024)7,076016
- **L.Qiu** et al., Dispersive analysis of $\eta(1405/1475)$ on the recent BESIII decay $J/\psi \rightarrow \gamma K_S^0 K_S^0 \pi^0$ (In progress)
- M.X.Duan, **L.Qiu**, X.Z.Ling and Q.Zhao, Predictions for feed-down enhancements at the $\Lambda_C \bar{D}$ and $\Lambda_C \bar{D}^*$ thresholds via the triangle and box singularities, PRD109(2024)3,L031507
- Y.Cheng, **L.Qiu** and Q.Zhao, On the widths of $\eta(1295)$ and $\eta(1405/1475)$, arXiv: 2302.01210
- Y.Cheng, **L.Qiu** and Q.Zhao, Understanding the non-trivial isoscalar pseudoscalar structures in the $K_S K_S \pi^0$ spectra in the J/ψ radiative decay, PRD111(2025)11,114015
- M.X.Duan, C.Gong, **L.Qiu** and Q.Zhao, J/ψ photoproduction near threshold and signals for the hidden charm pentaquarks, arXiv:2409.10364